

**PT. INDONESIA POWER
UNIT JASA PEMBANGKITAN PLTU JERANJANG
INTEGRATED MANAGEMENT SYSTEM**

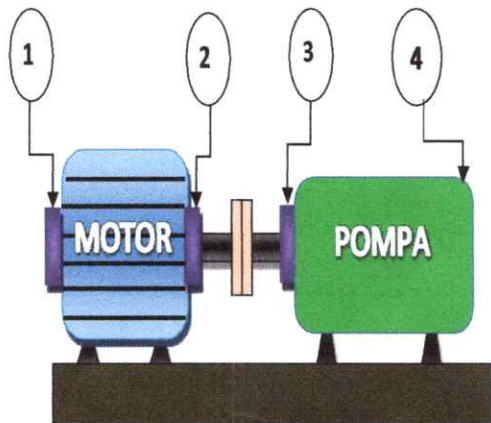
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Equipment : BOILER FEEDWATER PUMP 1A

KKS :
Technology : Vibrasi
Tanggal : 22 Mei 2020
Status : Kuning
Finding : Vibrasi Tinggi

I. Data Spesifikasi Motor

MOTOR		FAN	
Type, Mfg	: SY23372-1028, QINGJIANG	Type, Mfg.	: SHENYANG AST PUMP MANUFACTURING CO.,LTD
Speed	: 2980 rpm	Speed	: 2980 rpm
Power	: 1000	Capacity	: 140
Bearing Type	:	Bearing Type	:
Inboard Bearing	: NU 224EC	Inboard Bearing	: -
Outboard Bearing	: 6224/C3	Onboard Bearing	: -
Rotor Bar	: -	Total Blade	: -
Foundation	: Rigid		
FOTO EQUIPMENT		POINT PENGUKURAN	
			



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II. Hasil Pengukuran Vibrasi

Historikal pengukuran vibrasi BFP 1A

BULAN	VIBRASI BFP 1A																	
	MOTOR						COUPLING						POMPA					
	1V	1H	1A	2V	2H	2A	3V	3H	3A	4V	4H	4A	5V	5H	5A	6V	6H	6A
January-20	0.63	1.1	3.01	0.85	0.93	2.92	0.67	0.84	0.98	0.97	0.83	1.17	2.08	2.24	1.52	2.56	4.31	1.81
February-20	1.47	1.21	5.74	1.03	1.04	3.01	1	1.16	1.21	1	1.09	1.47	1.97	2.09	1.85	3.05	4.58	2.68
March-20	0.44	1.19	3.39	0.8	1.11	2.7	0.9	1.06	1.26	0.94	1.01	1.3	1.75	4.52	1.5	4.12	8.4	2.87
April-20	1.1	1.36	3.76	0.9	1.12	2.71	1.4	1.56	1.76	1.44	1.51	1.8	1.8	4.63	1.62	4.08	7.66	2.61
May-20	1.52	1.36	1.39	0.74	1.07	2.4	0.69	0.99	1.28	1.47	0.99	1.36	1.47	3.68	1.38	4.63	10.55	3.22

STANDART ISO 10816-3 (GROUP 1 RIGID)

	≥0	≥ 2.3	≥ 4.5	≥ 7.1
ALARM LIMIT	A	B	C	D

BFP 1A

CONTROL VALVE	VIBRASI (mm/s)																	
	BEARING MOTOR						BEARING COUPLING						BEARING POMPA					
	1V	1H	1A	2V	2H	2A	3V	3H	3A	4V	4H	4A	5V	5H	5A	6V	6H	6A
	1.86	2.06	3.28	1.9	2.47	5.38	0.61	1.52	1.23	0.69	1.43	0.53	0.4	3.16	0.46	0.82	4	0.56
	1.35	2.37	2.67	1.11	1.64	2.18	0.65	1.49	0.79	0.75	1.3	0.75	0.47	2.88	0.43	0.88	3.4	0.54
30%	0.78	1.73	1.87	0.91	1.3	2.5	0.65	1.19	1.63	0.71	1.24	1.23	3.75	9.68	2.37	10.36	8.37	7.77
50%	1.24	1.21	3.68	0.93	0.85	2.65	0.79	1.27	1.42	0.87	1.42	1.66	3.02	9.55	2.61	9.47	19.71	7.5

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Hasil Rekaman Spectrum Vibrasi :

✓ **Pada Motor :**

- Pada Bearing 1 (NDE Motor) overall vibrasi masuk status prewarning, dengan nilai tertinggi 3.68 mm/s RMS

✓ **Pada Pompa :**

- Pada Bearing 6 (NDE Pompa) overall vibrasi masuk status ALARM, dengan nilai tertinggi 19.71 mm/s RMS.

Kemungkinan :

✓ **Pada Motor :**

- Lakukan monitoring temperature dan current motor

✓ **Pada Pompa :**

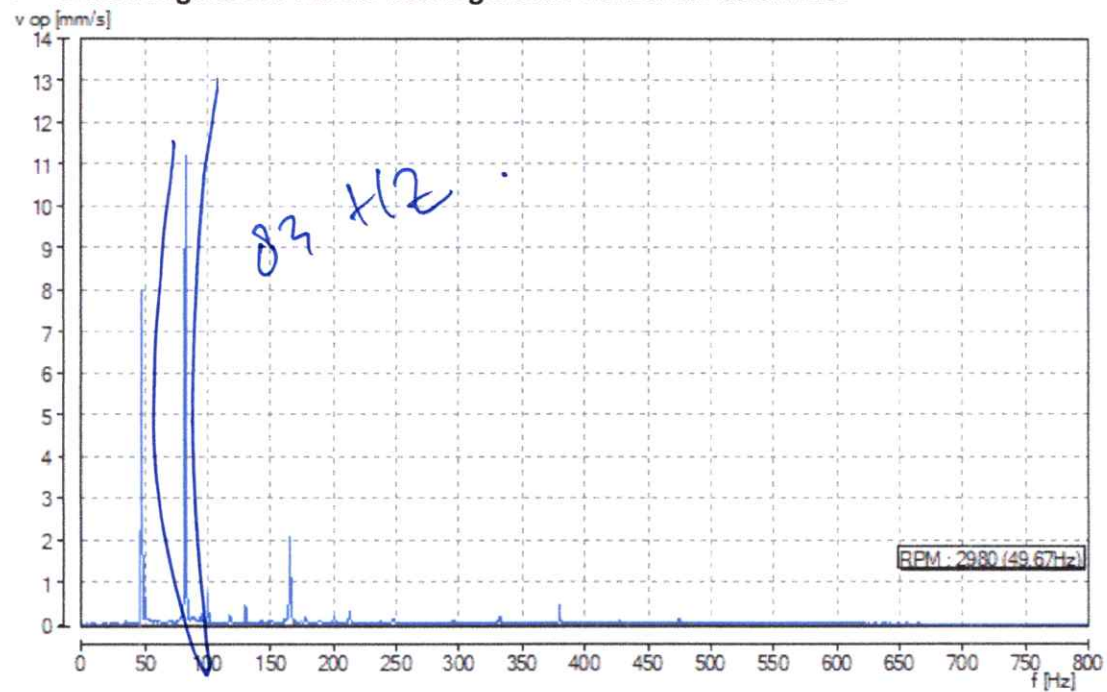
- Lakukan penguatan structure

✓

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➤ **Data Pengukuran Vibrasi Bearing 6 arah horizontal bulan mei**



Main Cursor = Not Active
 Delta Cursor = Not Active
 RPM = 2980
 RMS = 9.92

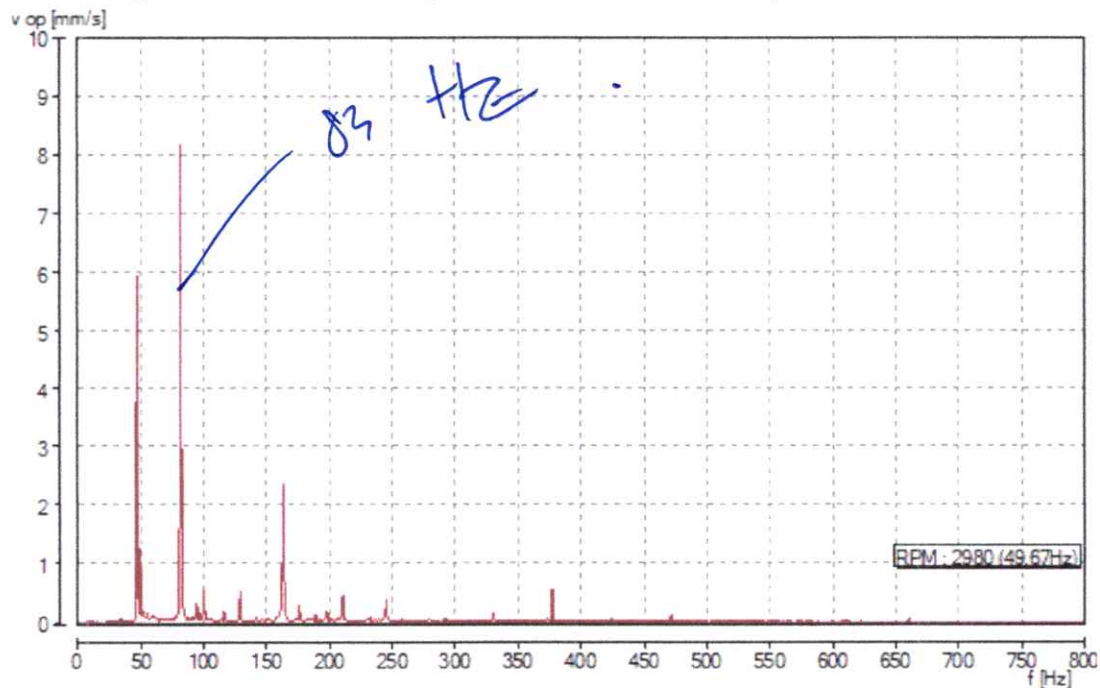
Channel #	A	
	X [Hz]	Y [mm/s]
1	82.50	11.23
2	47.50	7.99
3	165.00	2.09
4	49.50	0.87
5	100.00	0.73
6	130.00	0.47
7	380.00	0.45
8	212.50	0.32
9	95.00	0.28
10	117.50	0.24

✓

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➤ **Data Pengukuran Vibrasi Bearing 6 arah horizontal bulan April**



Main Cursor = Not Active
 Delta Cursor = Not Active
 RPM = 2980
 RMS = 7.47

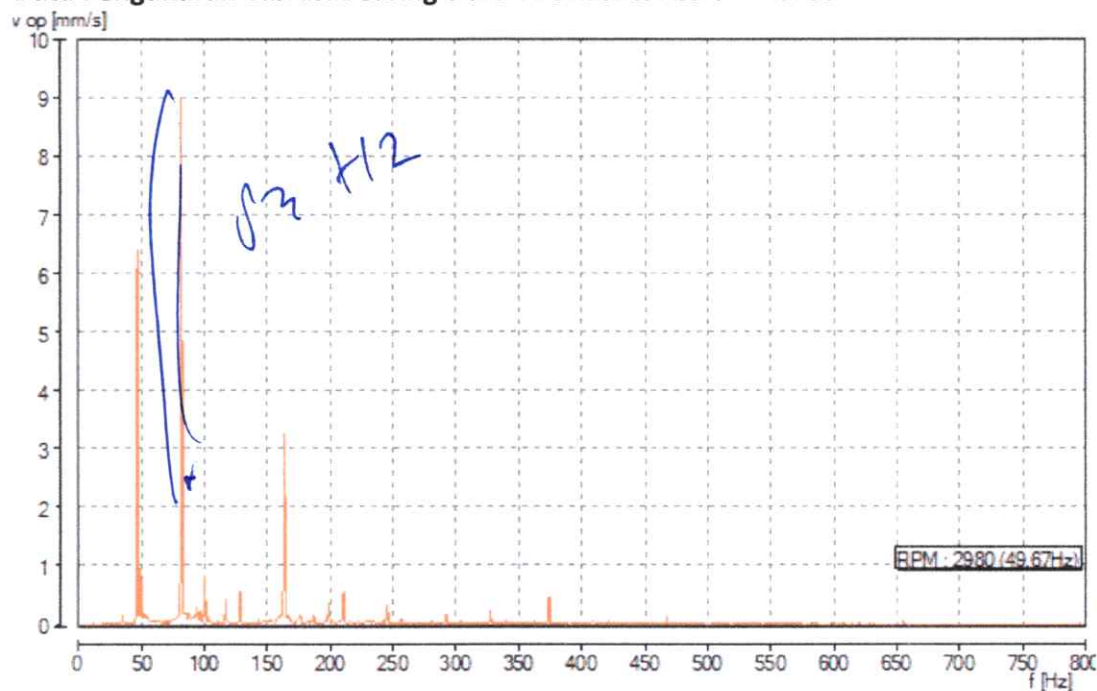
Channel #	A	Y [mm/s]
1	X [Hz]	
2	81.50	8.18
3	47.00	5.95
4	163.50	2.36
5	49.50	1.25
6	377.50	0.57
7	100.00	0.55
8	129.00	0.52
9	210.50	0.47
10	245.00	0.40
	94.50	0.33

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➤ **Data Pengukuran Vibrasi Bearing 6 arah horizontal bulan Maret**



Main Cursor	= Not Active
	= Not Active
Delta Cursor	= Not Active
	= Not Active
RPM	= 2980
RMS	= 8.26

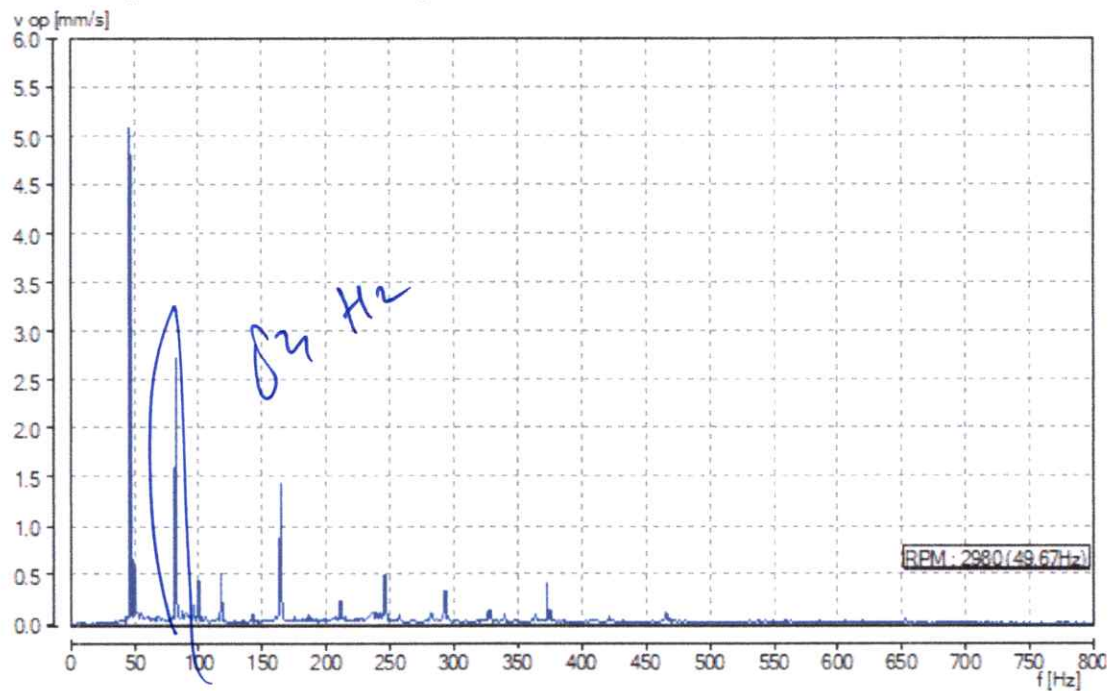
Channel #	A	
	X [Hz]	Y [mm/s]
1	82.00	9.01
2	47.00	6.41
3	163.50	3.23
4	49.50	0.96
5	100.00	0.82
6	128.50	0.59
7	210.50	0.57
8	374.50	0.48
9	117.00	0.42
10	198.50	0.34

✓

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➤ **Data Pengukuran Vibrasi Bearing 6 arah horizontal bulan Februari**



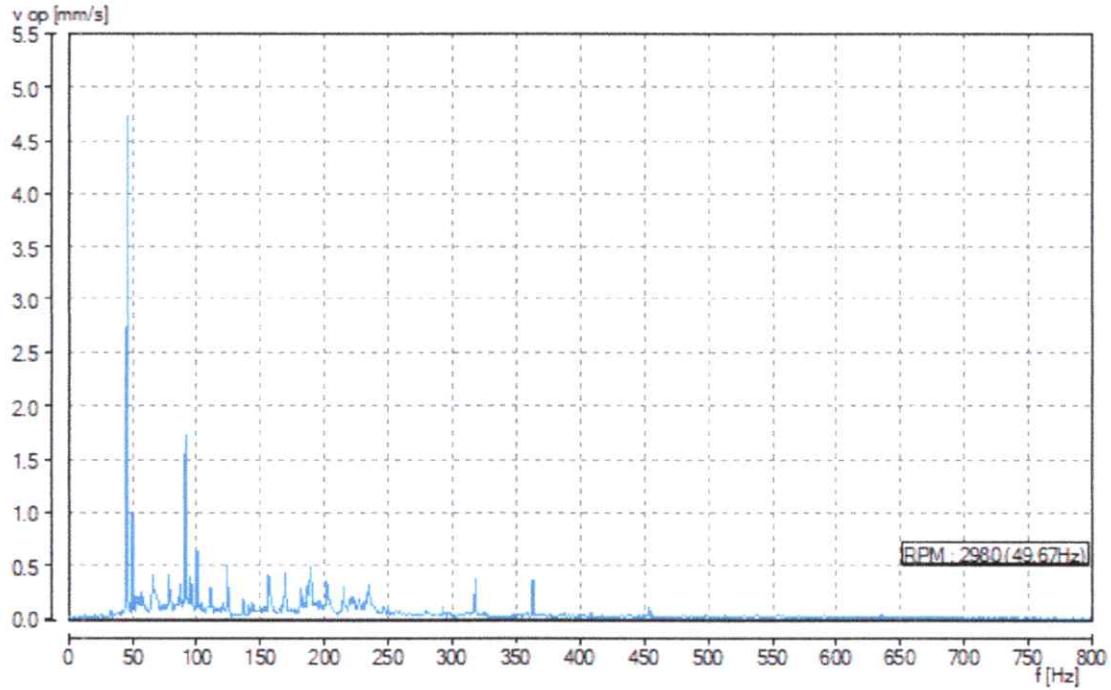
Main Cursor = Not Active
 Delta Cursor = Not Active
 RPM = 2980
 RMS = 4.33

Channel #	A	
	X [Hz]	Y [mm/s]
1	46.50	5.09
2	82.00	2.72
3	164.00	1.44
4	49.50	0.66
5	117.50	0.51
6	246.00	0.50
7	100.00	0.43
8	373.00	0.42
9	293.00	0.35
10	211.00	0.24

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➤ **Data Pengukuran Vibrasi Bearing 6 arah horizontal bulan Januari**



Main Cursor = Not Active
 Delta Cursor = Not Active
 RPM = 2980
 RMS = 3.91

Channel #	A	
	X [Hz]	Y [mm/s]
1	45.50	4.74
2	91.00	1.73
3	49.50	1.00
4	100.00	0.63
5	123.50	0.51
6	189.00	0.49
7	169.00	0.43
8	78.00	0.42
9	65.50	0.42
10	156.00	0.42

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III. Analisa :

- Pambatan Frek Natural
- Soak Rpm up ~~frek~~ ^{kecepatan} mulai 1/6 turn
artinya frek rpm BFP melewati frek natural.

IV. Kesimpulan :

- Rekejan dan menambahkan master
menekan frek natural pada BFP

V. Rekomendasi :

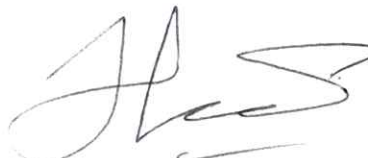
- pergunakan support pada ols atau
- menambahkan master pada pompa ~~(damping)~~
- menambahkan damping.

JERANJANG, 22 Mei 2020

MENGETAHUI


RICKY RINALDI
SPS RSO

DISUSUN OLEH


HERMAWAN
PS PdM